ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ME ENGINEERING — SEMESTER 3

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Derive Steady Flow Energy Equation (SFEE). Apply SFEE to Nozzle, Turbine, Compressor. [14 Marks]

[OR]

Q.1 (Alt) State First and Zeroth Laws of Thermodynamics. Calculate polytropic compression work and heat. [14 Marks]

--- UNIT II ---

Q.2 State Kelvin-Planck and Clausius statements of Second Law. Derive Carnot efficiency. [14 Marks]

[OR]

Q.2 (Alt) State Clausius Inequality. Prove Principle of Increase of Entropy for isolated systems. [14 Marks]

--- UNIT III ---

Q.3 Define Availability, Exergy, and Irreversibility. State Gouy-Stodola theorem. [14 Marks]

[OR]

Q.3 (Alt) Derive Maxwell's Thermodynamic Relations and Clausius-Clapeyron equation for phase change. [14 Marks]

--- UNIT IV ---

Q.4 Explain Steam formation with P-v-T diagrams. Describe Throttling Calorimeter for dryness fraction. [14 Marks]

[OR]

Q.4 (Alt) Explain Rankine Cycle with Reheat and Regeneration on T-s and Mollier diagrams. [14 Marks]

--- UNIT V ---

Q.5 Derive air-standard efficiency and MEP for Otto and Diesel cycles with P-v diagrams. [14 Marks]

[OR]

Q.5 (Alt) Compare Otto, Diesel, and Dual cycles for same compression ratio. Explain Brayton cycle. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****